ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

DIPLOMA (CSE)

Semester: IV

SUBJECT CODE: DCS403

COMPUTER ORGANIZATION

Time: 03:00 Hrs. **Maximum Marks: 60**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. **(12 Marks)**

Q. 1. Define the following terms: **(1×6 = 6 Marks)**

(i) Digital Computer.

(ii) Bus Arbitration.

(iii) Normalization.

(iv) Instruction.

(v) Interrupts.

(vi) Memory.

Q. 2. Answer the following questions: **(1×6 = 6 Marks)**

(i) What are different types of buses?

(ii) Write the concept of the stored program architecture.

(iii) What do you understand by special purpose register?

(iv) What is the role of an I/O interface in computer systems?

(v) What is the need for cache memory in computer systems?

(vi) Write the concept of virtual memory.

SECTION – B

NOTE:Attempt all questions. **(6×2×2 = 24 Marks)**

Q.3(i) (a) Describe the concept of a digital computer generation and its classification.

OR

(b) Explain the functional units and their interconnections in a computer system.

(ii)(a) How are signed numbers represented in computers using 2's complement? Explain with an example.

OR

(b) Discuss the process of Booth's multiplication algorithm with an example.

Q.4 (i)(a) What are the different addressing modes in a processor? Explain with examples.

OR

(b) Explain the concept of instruction formats in a computer system.

(ii)(a) What are interrupts? Discuss the types of interrupts in a computer system.

OR

(b) Explain the different modes of data transfer, including Programmed I/O, Interrupt-initiated I/O, and DMA.

SECTION – C

NOTE: Attempt all the questions. **(12×2 = 24 Marks)**

Q.5 (a) Explain the basic concept of memory hierarchy with examples.

OR

(b) Explain the concept of virtual memory and its implementation in modern computer systems.

Q.6 (a) What is Instruction Cycle? Explain types of instructions with example.

OR

(b) Describe the working principle of Direct Memory Access (DMA) and its advantages over programmed I/O.